Week1

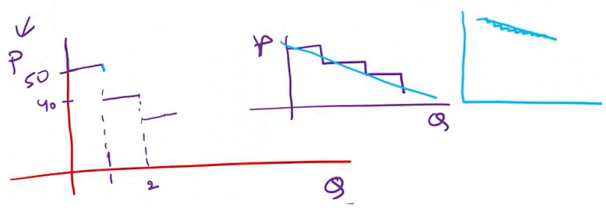
- Economics is a study of allocating scarce resources to satisfy *unlimited* individual wants or desires. Unless resources are scarce, there's no reason to study economics.

- Economics is an applied course in constraint optimization.
- In microeconomics we study:
  - Producers/Firms managed by Managers
  - Consumers
  - Markets
- Firms take low value inputs (raw materials) and produce high-value outputs.
- Exogenous Vs Endogenous variables: Prices of raw materials is exogenous, and how much of each should be bought is endogenous.
- While manager of the firm is interested in maximising profit or minimizing cost, customer is interested in maximising satisfaction/utility. In general, this is called constrained optimization, and mathematically represented as

$$\max_x \ f(x) \quad \text{s.t.} \quad g(x) < c$$

- In a perfectly competitive market, price of a commodity is decided by the market. In a monopoly too, it's decided by the seller, but constrained by the market demand curve.
- In a transaction, the following entities are involved - buyer, seller, environment and government. Note that the last two could influence the price and hence the profit.
- Buyer will purchase the good only when his/her value $BV \geqslant Price$. Seller will sell the good only when $Marginal\ Cost \leqslant Price$. Thus, $BV \geqslant Price \geqslant Marginal\ Cost$. This results in a value-add to the society that can be quantified as $BV - Marginal\ Cost$. This quantity is called $Total\ Surplus$, which are divided between buyer and seller, as *consumer surplus* and *producer surplus* respectively.
- Buyer is said to be rational, if he is able to make decisions without confusion, and exhibits consistency in his decisions (provided the environment doesn't change).
- Buyer's reservation price
  - The maximum price equal to their marginal willingness to pay for one additional unit.
  - As consumption increases, reservation prices typically fall. This is because each additional unit provides less extra satisfaction (law of diminishing marginal utility). This is the **utility-side view**.
  - As the market price decreases, buyers demand more units, since lower prices make additional units worthwhile to purchase. This is the **market-side view**.
  - If the market price is above the reservation price for a unit, the buyer will not demand that unit.
- Consumer Surplus

  - The difference between the buyer's reservation price (maximum willingness to pay) and the actual market price paid.
  - For example, if the market price of an item is Rs.50 and the buyer's reservation price is Rs.60, then consumer surplus = Rs.60 − Rs.50 = Rs.10.
  - The buyer seeks to maximize the utility gained from consumption, which translates into maximizing consumer surplus. This is the **utility-side view.**
  - Consumer surplus can be expressed in absolute terms (monetary difference) or in percentage terms (relative to the reservation price or expenditure).
- Demand (Quantity) is a function of price. As the price reduces, demand increases. This can be captured in a graph, plotted with Q on X-axis and P on Y-axis. This is called demand curve.



Note that demand curve is typically a downward sloping graph.

- The price above which demand is zero is called the choke price. In the above example, choke price = Rs.50/-
- Consider the data of the an apartment owner trying to rent out all his 10 apartments.

Price (Monthly Rental in thousand)= P	Quantity (Number of Apartment Rented)
P > 70	0
65 < P ≤ 70	1
60 < P ≤ 65	2
55 < P ≤ 60	3
40 < P ≤ 55	4
35 < P ≤ 40	5
30 < P ≤ 35	6
22 < P ≤ 30	7
20 < P ≤ 22	8
15 < P ≤ 20	9
P ≤ 15	10

Demand curve for this data can be represented as a linear graph $Q-1=-\frac{9}{55}(P-70)$. Note that $(70,1)$ and $(15,10)$ are points in the line. Above equation can be simplified to $Q=\frac{685-9P}{55}$.

- $Total\ revenue = QP = \frac{P(685-9P)}{55} = \frac{685P-9P^2}{55}$. Now, to maximize the total

reveue, we differentiate with respect to P and equate to 0, when we get

$P = \frac{685}{18} = 38.05 \approx 38.$

- Above calculations work as far it's a monopolistic market, and there're no other sellers in the market.
- Seller is interested in maximising the profit($\prod$). In mathematical terms,

  $\prod = Total\ revenue - Total\ cost = P.Q - C(Q)$, where P is the unit price, Q is quantity, and C(Q) is cost as a function of quantity.
- Seller's reservation price
  - The minimum price at which the seller is willing to supply a unit, equal to the marginal cost of producing that unit.
  - As output increases, reservation prices typically rise. This is because each additional unit typically costs more to produce. This is the **cost-side view**.
  - As price increases, suppliers supply more. This is because higher prices cover higher marginal costs and make additional units profitable. This is the **market-side view**.
  - If the market price is below the reservation price, the seller will not supply that unit.
- Producer Surplus
  - The difference between the market price a seller actually receives and the seller's reservation price (minimum acceptable price, equal to marginal cost).
  - For example, if the market price of an item is Rs.50 and the seller's reservation price for that unit is Rs.40, then producer surplus = Rs.50 − Rs.40 = Rs.10.
  - The seller seeks to maximize profit, which translates into maximizing producer surplus across all units supplied. This is the **cost-side view.**
  - Producer surplus can be expressed in absolute terms (monetary difference) or in percentage terms (relative to cost or revenue).
- Demand curve is plotted between *Quantity purchased by consumers* on X-axis and *Price per unit* on Y-axis. It has negative slope, and reducing slope as the quantity increases. At higher prices, consumers buy less; at lower prices, they buy more.
- Supply curve is plotted between *Quantity supplied by firms* on X-axis and *Price per unit* on Y-axis. It has positive slope, and increasing slope as the quanitity increases. At higher prices, produces are willing to sell more; at lower prices, they sell less.

- Problems

  1. Consider an apartment owner who has 4 apartments to be rented. There are 8 buyers who wishes to rent out an apartment. Their reservations prices (in 1000's of Rs) are shown in the following table.

Buyer	A	B	C	D	E	F	G	H
Res. Price	10	50	40	30	15	20	10	5

Rent can only be set in multiples of 5. Now, answer the following questions:

- What will be the price corresponding to the maximum revenue for the apartment owner?
- What will be the revenue in this case?
- What will be the price when all four apartments are rented?
- What will be the corresponding revenue when all four apartments are rented?

To answer these questions, firstly sort the reservation prices in desending order, and identify the respective buyers. Test candidate rents and make the following table.

Rent	Eligible buyers (≥ rent)	Apartments rented	Revenue
5	All 8 buyers	4	5 × 4 = **20**
10	A, B, C, D, E, F, G	4	10 × 4 = **40**
15	B, C, D, F, E	4	15 × 4 = **60**
20	B, C, D, F	4	20 × 4 = **80**
25	B, C, D	3	25 × 3 = **75**
30	B, C, D	3	30 × 3 = **90**
35	B, C	2	35 × 2 = **70**
40	B, C	2	40 × 2 = **80**
45	B	1	45 × 1 = **45**
50	B	1	50 × 1 = **50**

For example, if the rent is fixed at Rs.5K, all 8 buyers will be able to take it, since all their reservation prices are more than (or equal to) 5K. In this case, all 4 apartments will be taken.
As another example, if the rent is fixed at say 25K, only 3 buyers (B, C, D) will be able to take it, since only their reservation prices are more than (or equal to) 25K. In this case, only 3 apartments will be taken.

Thus, all above questions can be answered from the above table, like below.

- What will be the price corresponding to the maximum revenue for the apartment owner? Ans.30K
- What will be the revenue in this case? Ans.90K
- What will be the price when all four apartments are rented? Ans.20K

  - What will be the corresponding revenue when all four apartments are rented? Ans.80K

Week2

- $1\ net\ 10\ 30$ means the buyer gets 1% discount, if he pays the supplier within 10 days; the maximum period of payment is 30 days from purchase.
- *Discounts* are upfront price reduction on purchase of goods, and *Rebates* are reimbursements upon full payment.
- *Returns* are goods sent back by the buyer to the seller, due to defects, damage, wrong shipment or unsold stock under certain agreements.
- *Allowances* are money or concessions given by a manufacturer/wholesaler to a retailer for prominent display or promotion of goods
- *Opportunity cost* is what a seller/producer/firm gives up to use a particular resource. In other words, it's the value of the *next best alternative forgone* when a resource is used for particular purpose. For example, If a factory uses steel to make cars, the opportunity cost is the profit it could have earned by making trucks instead. Note that resources can be transactional or non-transactional (owner-supplied).
- Transactional cost is typically called *explicit cost* and non-transactional cost is called *implicit cost*. Transactional cost is almost always money, and examples for non-transactional cost are time, rent paid for the facilities.
- Accounting doesn't consider implicit costs, but economists do. Thus,

$$Accounting\ Profit = Revenue - Explicit\ costs$$
$$Economic\ Profit = Revenue - (Explict\ costs\ +\ Implicit\ costs)$$

- Suppose A and B run identical grocery stores. Both earn a profit of 5K. A owns the store space. If A rents out his store space, he'll earn 5K. However, B pays a rent of 5K. What is the accounting/economic profit of A and B?
  - A earns 5K profit. So, his accounting profit is 5K. But, since he's foregoing a rent of 5K, his economic profit = 0
  - B earns 5K profit. So, his accounting profit is 5K. Since there's no implicit cost, his economic profit = 5K.
- Sunk cost is one that have already been incurred and cannot be recovered. It shouldn't be considered while making any future decisions and hence should be ignored, as per economists. However, accountants treat such expenditures as capitalized assets and then allocate them over time through depreciation or amortization. Such expenditures will continue to appear in financial reporting, until written off.
- Consider implicit cost and ignore sunk cost while making a business decision.
- When cost depends on quantity, it's called variable cost; when it's independent of quantity, it's called fixed cost. Mathematically, this can be written as follows.

$Total\ cost = C(Q) = FC + VC(Q)$

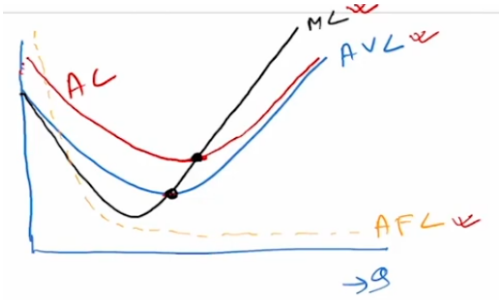
- Dividing the above equation by Q, we get

  $Average\ Cost = \frac{C(Q)}{Q} = \frac{FC}{Q} + \frac{VC(Q)}{Q} = AFC(Q) + AVC(Q)$

  Note that *AFC* is a function of *Q*, though FC is independent of Q. On the other hand, *AVC* can be independent of Q in certain cases, though VC is always dependent on Q.
- *AFC* always reduces with increasing quantity. *AVC* also typically reduces with increasing quantity, but can increase or even stay unchanged. If *AVC* increases with *Q*, *AC* (being a sum of AFC and AVC), decreases initially and then increases, and has a U-shape. If *AVC* decreases with *Q*, *AC* (being a sum of AFC and AVC), continously decreases. If *AVC* is unchanged, *AC* (being a sum of AFC and AVC), decreases initially and then asymptotically approaches the constant AVC level.
- Marginal cost is the cost to produce on more unit of good. Mathematically, $MC(Q) = \frac{C(Q+1) - C(Q)}{(Q+1) - Q}$. It can also be defined as $MC(Q) = \frac{C(Q) - C(Q-1)}{Q - (Q-1)}$. Note that these two definitions might not yield the same value for *MC*.
- It implies that *MC* can vary with quantity, and should be ideally defined using calculus. Thus, in general, $MC(Q) = \frac{C(Q+h) - C(Q)}{(Q+h) - Q}$, where $h \to 0$. In other words, the rate of change of *Total cost* (represented graphically by a tangent to the cost curve at *Q*) is called *Marginal cost*.
- Relation between MC and AC
  - When $MC > AC$, it pulls *AC* upwards. For example, if it takes Rs.100/- to produce 10 items. and takes Rs.111/- to produce 11 items, $MC = Rs.11/-$.

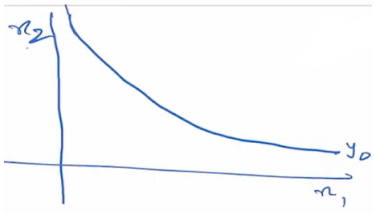
    Now, in the first case, $AC = \frac{100}{10} = 10$, and in the second case,

    $AC = \frac{111}{11} = 10.09.$
  - When $MC < AC$, it pulls *AC* downwards.
  - When $MC = AC$, AC is the lowest.
- Here is the graph depicting AC, AVC, MC and AFC.

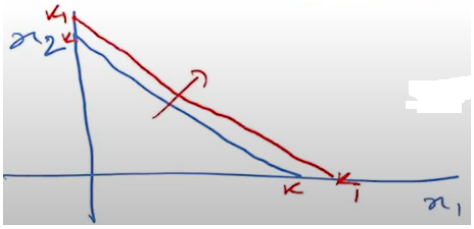


- Here is the general rule.

  - If the latest unit (MC) costs less than the average, it pulls the average down.
  - If the latest unit costs more than the average, it pushes the average up.
  - If it equals the average, the average stays flat (minimum point)
- Using calculus, we can prove that $\frac{d(AC)}{dQ} = \frac{1}{Q}[MC - AC]$. Thus, when $MC > AC$, the slope of the *AC* curve increases. When $MC < AC$, the slope of the *AC* curve decreases. When $MC = AC$, the AC curve remains unchanged.
- A firm takes n inputs and produces m outputs, and can be mathematically represented as $y = f(x1, x2, x3...xn)$. Note that y is vector consisting of m elements; ie $y = (y1, y2, y3...ym)$. To make things simpler, we often assume there are two inputs *L (labor)* and *K (capital),* and that y is a scalar.
- Different kinds of production function
  - Fixed proportions (Leontief): Each unit of output is produced by L and K in the ratio l:k. THis can be represented mathematically as $min\left(\frac{x1}{l}, \frac{x2}{k}\right)$, where x1 is the number of units of L employed, and x2 is the number of units of K employed. Note that *x1* and *x2* are assumed to be greater than 0, else $y = 0$ (which is disallowed). No substitution is allowed.
  - Variable proportions:
    * Cobb Douglas: Mathematically represented as $y = x_1^a x_2^b$, where *x1* is the number of units of L employed, and *x2* is the number of units of K employed. Note that it's assumed that $a > 0 \; and \; b > 0$. If *a = 1,* and *b = 1*, $y = x1x2$.
    * Linear: Mathematically represented as $y = x1 + x2$, where where *x1* is the number of units of L employed, and *x2* is the number of units of K employed. Allows perfect substitution between inputs.
- In Cobb Douglas and Linear production, it's possible to attain maximum output by using different amount of input resources, unlike in Leontief production. In the latter, it's mandatory to have l units of L and k units of K to produce output.
- An isoquant is often used to represent an efficient production function. Thus, to represent $y = x_1x_2$, we can use a hyperbola, all points of which represent equal output.
- Shown below are the isoquants for various production functions
  - Cobb-Douglas.



– Linear



– Leontief



Note: In the above case, it's assumed y = 10, for the sake of drawing the figure.

- Average product due to labor $APL = \frac{y}{L}$, assuming that K is constant. Similarly, average product due to capital $APK = \frac{y}{K}$, assuming that L is constant.
- Marginal product due to labor $MPL = \frac{\partial y}{\partial L}$. Marginal product due to capital $MPK = \frac{\partial y}{\partial K}$.
- Marginal product decreases as more output is produced, since efficacy of input resources reduces as more and more of it is employed. This is called law of diminishing marginal product.
- Legal organization of firms - Sole proprietorship, Partnership, Limited liability, Corporation. The first two have unlimited liability for loans, the last two have limited liability. The first two don't require registration, and last two need registration. The first three are taxed only once, but the last one is taxed twice.

- To recall from Week1,Profit $\prod = Total\ revenue - Total\ cost = P.Q - C(Q)$, where P is the unit price, Q is quantity, and C(Q) is cost as a function of quantity. If quantity produced Q_0 increased by 1 to $Q_0 + 1$, *Total Revenue (TR)* increases by *Marginal Revenue (MR)*. At the same time, *Total Cost (TC)* increases by *Marginal Cost (MC)*.
- For profit to rise when producing an extra unit, $MR > MC$. When $MR = MC$, seller has attained maximum profit, and wouldn't want to produce more output. Note that this doesn't work in the case where units produced are discrete, like in the train example in the *Problems* section below. In this case, profit maximization occurs when $MR(Q_0) > MC(Q_0)$ and $MR(Q_0 + 1) < MC(Q_0 + 1)$
- Problems
  - To find how to decide what to study to maximize marks:

No. of Hours of Study	Economics	Gain in Marks for each additional hour on Economics	Statistics	Gain in Marks for each additional hour On Statistics
0	7.0		6.0	
1	7.7	7.7 - 7 = 0.7	6.8	6.8 - 6 = 0.8
2	8.2	8.2 - 7.7 = 0.5	7.4	7.4 - 6.8 = 0.6
3	8.5	8.5 - 8.2 = 0.3	7.8	7.8 - 7.4 = 0.4

    Note that per marginal utility calculated for each additional hour spent, we could find that the first hour should be spent on statistics, second hour on economics and third spent again on statistics.
  - To find how many trains to run, given the benefit obtained and cost required to run trains.

Number	Daily Total Benefit in Lakhs	Daily Average Benefit (in lakhs/Train)	Marginal Benefit(in Lakhs/Train)	Marginal Cost (in Lakhs/Train)
0	0	0		
1	30	30.0	30 (30-0)	15
2	48	24.0	18 (48-30)	15
3	60	20.0	12	15
4	64	16.0	4	15
5	66	12.5	2	15

  - A firm faces price P = 66, independent of its output Q. The cost function is given as $\frac{Q^3}{3} - 10Q^2 + 102Q + 3$. WHat is the most profitable level of output?
    Ans:We use the formula $MR = MC$.
    $MR = P(Q+1) - PQ = P.$

$$MC = \frac{dC}{dQ} = Q^2 - 20Q + 102.$$

Equating *MR* and *MC*, we get

$$Q^2 - 20Q + 102 = 66 \implies Q^2 - 20Q + 36 = 0 \implies Q = 2 \; or \; Q = 18$$

Now, taking double-derivative of profit, $\frac{d\left(66 - Q^2 + 20Q - 36\right)}{dQ} = -2Q + 20.$

At $Q = 2$, double-derivative is positive, and at $Q = 18$, double-derivative is negative. Thus, the maximum profit is attained at $Q = 18$.

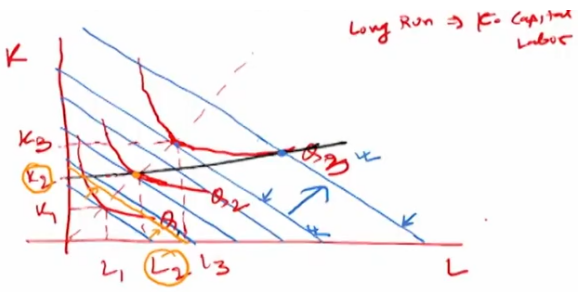
Week3

- Marginal productivity is the derviative of production curve (total product curve). It tends to reduce as quantity (X-axis) increases. This happens because additional units of input have less and less of the fixed factors (like machines, land, or space) to work with, and each extra worker or machine adds less to total output than the previous one. This is termed diminishing marginal productivity.
- If the production function is $y = f(L, K)$, where L is labor and K is capital, then the total differential of output is $dy = \frac{\partial f}{\partial L}dL + \frac{\partial f}{\partial K}dK$. Here $\frac{\partial f}{\partial L}$ is called marginal product of labor (MPL) and $\frac{\partial f}{\partial K}$ is called marginal product of capital (MPK). *dL* and *dK* are small changes in labor and capital. Thus, if you add a little more labor (*dL*), output rises by $MPL{*}dL$. If you add a little more capital (*dK*), output rises by $MPK{*}dK$
- Note that *MPL* initially increases, but reduces as more and more quantity is produced. The point at which decline of *MPL* occurs can be found by taking the double-derivative of the production function with respect to *L*, $\frac{\partial^2 f}{\partial L^2} = 0$. The same phenomenon occures with MPK too.
- Isoquant shows the trade-off between L and K, and is called Marginal Rate of Technical Substitution (*MRTS)*. *MRTS* is defined as how much capital you can give up, if you add one more unit of labor while keeping output constant.
- For example, if $K = 6, L = 2$ would produce *12* units of product, $K = 5$ requires $L = \frac{12}{5}$ to remain in the same isoquant. Here, when a unit of K is reduced , L must be increased by $\frac{12}{5} - 2 = \frac{2}{5}$. This essentially implies that one unit of capital is substituted by $\frac{2}{5}$ units of labor. This is termed as Marginal Rate of Technical Subsitution (MRTS). However, note that at different places along the isoquant, *MRTS*

differs. In other words, slope along the isoquant represents the *MRTS*.

- By convention, we say MRTS is the rate of substituting L for K $\left(\frac{dK}{dL}\right)$. Here K is subsituted by L (L is subsituting for K). In general, $\frac{dy}{dx}$, where x and y are inputs in the production process.
- It can be proved that $MRTS = -\frac{MPL}{MPK}$
- In the case of Linear Production technology $y = ax_1 + bx_2$, where a and b are called production parameters, $x1\ and\ x_2$ are the inputs, $MRTS = \frac{a}{b}$.
- In the case of Cobb-Douglas function $y = x_1x_2$, where $x1\ and\ x_2$ are the inputs, $MRTS = \frac{x_2}{x_1}$.
- Assuming production function $y = f(x_1, x_2 \ldots x_n)$, and each of the inputs cost w_i per unit, the total cost of production is $\sum_{i=1}^{n} w_i x_i$
- Minimizing cost of production attempts to find $\min \sum w_i x_i$, such that $y = f(x_1, x_2 \ldots x_n)$
- If the production uses only one input x, cost of production can be given as $y = wf^{-1}(y_0)$, where y_0 is the quantity produced.
- If the production is linear and represented as $y = ax_1 + bx_2$ and uses two inputs $x_1\ (costs\ w_1)\ and\ x_2\ (costs\ w_2)$
  - In order to produce 1 unit of y, it takes $\frac{1}{a}$ unit of x_1 (costing $\frac{w_1}{a}$) or $\frac{1}{b}$ unit of x_2 (costing $\frac{w_2}{b}$)
  - If $\frac{w_1}{a} > \frac{w_2}{b}$, then input x_2 should be used. Else, use input x_1.
  - If y_0 units need to be produced, $Total\ cost = min\left[\frac{w_1}{a}, \frac{w_2}{b}\right] * y_0$
  - In other words, $x_2^* = \frac{w_1}{a}.y$ (when input1 is used), and $x_1^* = \frac{w_2}{b}.y$ (when input2 is used)
  - In this case, inputs are perfect substitutes, and firm choose cheaper ones.
- If the production is Leontief and represented as $y = min(ax_1, bx_2)$ and uses two inputs $x_1\ (costs\ w_1)\ and\ x_2\ (costs\ w_2)$

  - In order to produce 1 unit of y, it takes $\frac{1}{a}$ unit of x_1 (costing $\frac{w_1}{a}$) and $\frac{1}{b}$ unit of x_2 (costing $\frac{w_2}{b}$)
  - If y_0 units need to be produced, $Total\ cost\ = \left(\frac{w_1}{a} + \frac{w_2}{b}\right) y_0$
  - In this case, inputs are perfect complements and must be used in fixed proportions.
- If the production is Cobb-Douglas and represented by $y = x_1^{\alpha} x_2^{\beta}$ and uses two inputs $x_1\ (costs\ w_1)\ and\ x_2\ (costs\ w_2)$
  - Goal is to minimize cost $C = w_1 x_1 + w_2 x_2$
  - When cost is minimized, the isocost line of the input mix ($w_1 x_1 + w_2 x_2$) is tangential to the isoquant for y_0. In other words, the slope of isoquant and isocost line should be equal.
  - Slope of isoquant = $MRTS = -\frac{MP_1}{MP_2}$
  - Slope of isocost line = $-\frac{w_1}{w_2}$
  - Thus, at the minimized cost situation, $\frac{MP_1}{MP_2} = \frac{w_1}{w_2}$. THis can also be written as $\frac{MP_1}{w_1} = \frac{MP_2}{w_2}$.
  - Since 1 unit of x_1 costs w_1, it implies that $\frac{1}{w_1}$ unit of x_1 will cost $Rs.1/-$. Thus, if $Rs.1/-$ is spent on x_1, production will increase by $\frac{MP_1}{w_1}$. Similarly, if $Rs.1/-$ is spent on x_2, production will increase by $\frac{MP_1}{w_2}$
- Since the optimized input quantities x_1^* and x_2^* are are functions of $w_1,\ w_2,\ y$, minimized cost for the production turns out to be a function of $w_1,\ w_2,\ y$ too. This minimized cost is called *cost function*.
- This scenario is represented graphically as follows.



In other words, the optimal cost is achieved where the isocost and isoquant intersect.

- Problems

Example: Cobb–Douglas Production

Suppose:

- Production function:

$$y = x_1^{0.7} x_2^{0.3}$$

- (so $\alpha = 0.7, \beta = 0.3$, constant returns to scale since)
- Input prices: $w_1 = 2, w_2 = 8$
- Required output: $y = 100$

Step 1: Tangency condition

$$\frac{MP_1}{MP_2} = \frac{w_1}{w_2}$$

Marginal products:

$$MP_1 = 0.7 x_1^{-0.3} x_2^{0.3}, \quad MP_2 = 0.3 x_1^{0.7} x_2^{-0.7}$$

So:

$$\frac{MP_1}{MP_2} = \frac{0.7}{0.3} \cdot \frac{x_2}{x_1} = \frac{7}{3} \cdot \frac{x_2}{x_1}$$

Set equal to $w_1/w_2 = 2/8 = 1/4$:

$$\frac{7}{3} \cdot \frac{x_2}{x_1} = \frac{1}{4}$$

$$\frac{x_2}{x_1} = \frac{3}{28}$$

Step 2: Production constraint

We want output $y = 100$ with production function:

$$y = x_1^{0.7} x_2^{0.3}$$

From Step 1 we found the optimal ratio:

$$x_2 = \frac{3}{28} x_1$$

Substitute this into the production function:

$$100 = x_1^{0.7} \cdot \left(\frac{3}{28} x_1\right)^{0.3}$$

$$100 = x_1^{0.7} \cdot \left(\frac{3}{28}\right)^{0.3} \cdot x_1^{0.3}$$

$$100 = \left(\frac{3}{28}\right)^{0.3} \cdot x_1^{1.0}$$

Step 3: Numerical evaluation

$$\frac{3}{28} \approx 0.10714$$

Raise to power 0.3:

$$0.10714^{0.3} \approx 0.5119$$

So:

$$x_1 = \frac{100}{0.5119} \approx 195.4$$

Then:

$$x_2 = \frac{3}{28} \cdot 195.4 \approx 20.95$$

Step 4: Minimized cost

$$C^* = w_1 x_1 + w_2 x_2$$

$$C^* = 2(195.4) + 8(20.95)$$

$$C^* \approx 390.8 + 167.6 = 558.4$$

- Calculus-based methods can't be applied on a linear technology, since it'll lead to a *corner case*, where *derivative* isn't possible. Hence, use the input that yields the maximum productivity per dollar. Following is an example problem.
  Suppose that we have the linear production function $Q = 10L + 2K$. Price of labor w is $5 per unit and price of capital service r is $2 per unit. What is the optimal

combination of the inputs required to minimize the cost, if we need to produce 200 units of product.

Here, we can't apply the calculus technique that we used with Cobb-Douglas. $MP_L = 10;\ MP_K = 2.\ \frac{MP_L}{w} > \frac{MP_K}{r}$, so only labor will be used. Since we need to produce 200 units of product, $10L + 2K = 200$. Since no capital will be used, we could write the above equation as $10L = 200 \Longrightarrow L = 20$. Naturally, $K = 0$. This is the optimal combination of the inputs.

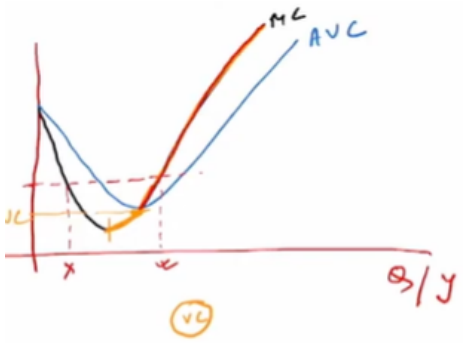
- When inputs are scaled up/down and output is also scaled up/down in same ratio, it's called Constant Return to Scale (CRS). Thus, when
  - $f(\lambda x_{10}, \lambda x_{20}) = \lambda f(x_{10}, x_{20})$, production is said to be in CRS.
  - $f(\lambda x_{10}, \lambda x_{20}) > \lambda f(x_{10}, x_{20})$, production is said to be in IRS.
  - $f(\lambda x_{10}, \lambda x_{20}) < \lambda f(x_{10}, x_{20})$, production is said to be in DRS.
- It's possible that *CRS, IRS and DRS* occur in different ranges of production. Typically
  - At low levels of production, firms often experience IRS (efficiency gains, specialization, indivisibilities)
  - At moderate levels, they may settle into CRS (balanced scaling)
  - At high levels, DRS can set in (coordination problems, resource constraints)
- Additivity and divisiblility holds in the case of CRS. Additivity implies that adding the same production process with itself will yield two times the output. Divisibility implies that a production process can be split, by bringing in the time factor. Thus, instead of producing half a car, you could produce one car in two days.
- To give an example for IRS, a box volume of $1m^3$ is obtained from an sheet area $6m^2$. But, a box volume of $8m^3$ can be obtained from an sheet area $24m^2$. Here, 4 times increase in sheet area (inputs) results in 8 times increase in box volume (output)
- To give an example for DRS, after reaching the maximum utilization of a hidden input, using any more of it could reduce the output.
- In the case of Cobb-Douglas,
  - if $\alpha + \beta = 1$, it results in CRS.
  - if $\alpha + \beta > 1$, it results in IRS.
  - if $\alpha + \beta < 1$, it results in DRS.
- Under CRS, total cost of inputs ($TC_2 = \lambda . TC_1$) and output ($y_2 = \lambda y_1$) are in the same ratio. In other words, the total cost increases at a constant rate. Hence average cost remains same as before scaling.
- Under IRS, total cost of inputs ($TC_2 = \lambda . TC_1$) rises less faster in comparison to output ($y_2 > \lambda y_1$). Hence average cost decreases.
- Under DRS, total cost of inputs ($TC_2 = \lambda . TC_1$) rises less faster in comparison to output ($y_2 < \lambda y_1$). Hence average cost increases.
- When total cost incurred to produce multiple items together is lesser compared to what

it takes to produced these items separately, it results in *returns to scope*.

- $\sum_{i=1}^{n} MC(i) = VC(n)$. Alternatively, through calculus it can be proved that $VC(Q) = \int_0^n MC dQ$. Note that Q varies from 0 to n.
- When Average Product of Labor (APL) is decreasing, Marginal Product of Labor (MPL) is less than APL. Similarly, when APL is increasing, MPL is greater than APL. When APL is at its peak, $MPL = APL$
- Problems
  - If production function is $Q = 10L + 2K$. Cost function $C = 5L + 2K$. Find the optimal combination of L and K, in order to produce *100* units. What is the minimum cost?
    Ans:Since the production function is linear, we can't use calculus. Instead, select the higher among $\frac{MPL}{w}$ and $\frac{MPK}{r}$. In this case, $\frac{MPL}{w}$ gives a higher value (2), in comparison with $\frac{MPK}{r}$ (1). So, optimal combination of inputs occurs when *L* replaces all of *K*. Substituting $K = 0$ into the production function, we get $200 = 10L \implies L = 20;\ K = 0$ is the optimal combination. Substituting $L = 20$ into the cost function, we obtain lowest cost as $C = 5 * 20 = 100$

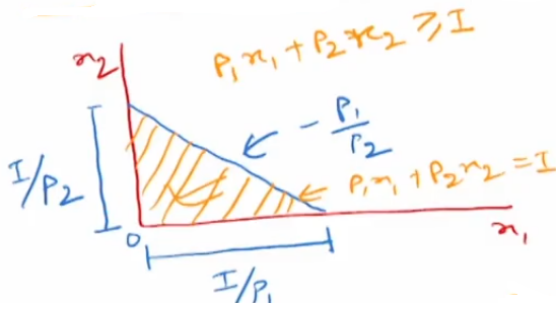
Week4

- As we saw previously, when $MC < AVC$, AVC reduces. When $MC > AVC$, AVC increases.

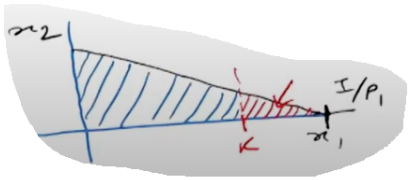


- A firm's supply curve is determined by its marginal cost curve. A firm will only supply a product if the market price for that product is at least equal to the marginal cost of producing it. Because the MC curve slopes upward, a higher price is required to induce a firm to supply more output.
- Upward-sloping part of the MC curve that lies above the AVC curve determines the supply function.

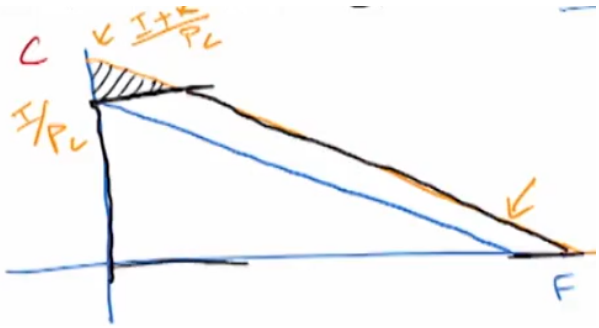
- The firm will only supply if the price is *at least* equal to the minimum AVC. If the price falls below this point, the firm loses more by staying open than by shutting down, so the supply becomes zero.
- Optimizing the product cost discussed during last week works only on the long-run, when both inputs can be changed simultaneously. In a *short run*, at least one factor of production (typically, capital) can't be changed.
- When K remains constant in the production process, the cost incurred to maintain the same level of production is *more* compared to when all factors can be changed, as can be noticed from the above graph. Thus, short-run cost is typically more than long-run cost.
- Assuming that capital K can't be changed in short-run, $marginal\ cost\ in\ short\ run\ = \frac{w}{MPL}$. When the capital *K* is higher, it complements labor and hence MPL increases, thus reducing the marginal cost in short run.
- Building blocks of consumer behaviour:
  - Consumption possibility set.
  - Consumption feasibility set
  - Preferences
  - Behavioral assumption.
- Consumption possibility set.
  - Represented by n-tuple, each of whose elements is a *non-negative* quantity that might be 'possibly' consumed by a user.
  - Consumption possibilities are typically represented in terms of bundles (represented using X), where the bundles follow the below properties
    * *additivity* implying $x \in X,\ y \in X \implies x + y \in X$
    * *divisibility* implying $x \in X \implies \alpha x \in X$, where $0 \leqslant \alpha \leqslant 1$.
  - Give two bundles $x,\ y \in X$ then any mixture of the two - say, $\lambda x + (1 - \lambda)y$ - must also be possible. Average mixtures are preferred, over extremes.
- Consumption feasibility set. Also considers the feasibility, including financial feasibility.
  - Legal, Geographical and financial constraints could be reason for infeasibile bundles.
  - $P_1 x_1 + P_2 x_2 \leqslant I$, where x_1, x_2 are quantities of goods in the bundle, P_1, P_2 are prices of the respective items and I is the income.
  - Below graph represents the budget line. The shaded part represents the consumption feasibility set, after subjecting to constraints.



- If all income were spent on x_1, you could buy $\frac{I}{P_1}$. This is the horizontal intercept. If P_1 increases (say, due to taxation), the horizontal intercept decreases.
- Similarly, If all income were spent on x_2, you could buy $\frac{I}{P_2}$. This is the vertical intercept. If P_2 increases, the vertical intercept decreases.
- If I decreases (due to taxation of income), both x and y intercepts decrease.
- $\frac{P_1}{P_2}$ is the exchange rate. This is the slope.
- If $x_1 \leqslant K$, it's called rationing and this will invalidate part of the budget set as shown below.



- If coupons worth K units are available, but can't be used to purchase x_2 and this will invalidate part of the budget set as shown below.



- Preferences. Prefer one over other, or indifferent to either.
  - $y \succ x$ means that the bundle y is preferred over bundle x.
  - $y \sim x$ means consumer is indifferent between the bundles y and x.
  - $y \succcurlyeq x$ means bundle y is at least as good as bundle x. It's a combination of $y \succ x$ and $y \sim x$.
  - Given bundles $x, y, z \in X$, it's assumed that, if $x \succcurlyeq y$ and $y \succcurlyeq z$, then $x \succcurlyeq z$. Alternatively, we say $U(x) \geqslant U(z)$, where U stands for utility function.
  - It's possible to rank bundles according to preferences of one over other, and given them numbers so that its easier to compare to each other. This works even with higher dimensional bundles.
- Behavioural assumption.
  - Monotonicity means, more the better. If $y \succcurlyeq x$ and has more of at least one good, then $y \succ x$
  - Local non-satiation is a weaker rule than monotonicity. For any bundle x, there is always a bundle y extremely close to it such that $y \succ x$
  - Monotonicity implies local non-satiation. But the reverse isn't true always. If you always prefer *more*, then you can always find a better bundle by just adding a tiny bit more of one good. Therefore, you are never *satiated*. However, it's possible for you to be non-satiated due to a *bad,* in which case *more* means *worse* and hence monotonicity isn't achieved.
  - If the consumer is perfectly satisfied with 2 *Fish* and 2 *Coconut* in a two-good world, the indifference curves are concentric circles with the *(2,2)* as the centre. Everywhere else, there's non-satiation. Now, same amount of indifference (non-satiation) occurs everywhere at equal distance from the center. Thus, at all points on a concentric circle with radius 10, same amount of indifference occurs, but increases when the circle's radius rise to 11.
  - If your perfect satisfaction is at (2, 2), your utility function is usually something like $U(x_1, x_2) = -\left[(x_1 - 2)^2 + (x_2 - 2)^2\right]$
  - Indifference curves generally satisfies the following rules.
    * Can't have positive slopes, if monotonicity was satified.

    * Can't have thick zones, if local non-satiation was satisfied.
    * Can't intersect with each other.
  - Indifference curves are downward sloping in generally. Typically, shape of the curve is convex, like the graph of $\frac{1}{x}$.
  - MRS is slope of indifference curve and used in the case of consumer. Mathematically, $MRS = \frac{MU_{x1}}{MU_{x2}}$. Note that MRTS is slope of isoquant and used in the case of producer.
  - If indifference curve is represented as $ax_1 + bx_2 = K$, (in other words, when *MRS* is a constant), then the inputs are called perfect substitutes. If indifference curve is represented as $min(ax_1, bx_2)$, then the inputs are perfect complements. If indifference curve is represented as ${x_1}^{\alpha}{x_2}^{\beta}$ (Cobb Douglas), then the inputs are substitutable, but the rate of substitution changes with the input quanties. If the utility function is represented as $x_1^{0.5} + y$ (quasi-linear), the preferences are linear in one good, but non-linear on the other.

Week5

- The consumer's problem is about picking the best bundle of goods from a feasible set that maximizes a consumer's satisfaction or happiness. Mathematically, it's represented as maximizing the utility function $U(x)$ subject to the budget constraint $Px \leq I$, where *P* is the price vector, *x* is the quantity vector, and *I* is the income.
- In order to find the optimum (maximum) utility, we must take a small amount of income ΔI from spending on x_1 and spend it instead on x_2. This implies that we're buying $\frac{\Delta I}{P1}$ fewer units of x_1 and $\frac{\Delta I}{P2}$ more units of x_2. Assuming that MU_1 and MU_2 are the marginal utilties of x1 and x2 respectively, this changes the total utility by $-\frac{\Delta I}{P1}.MU_1 + \frac{\Delta I}{P2}.MU_2$. This change in utility must be less than or equal to 0, else we've not achieved maximum total utility already. Thus, $-\frac{\Delta I}{P1}.MU_1 + \frac{\Delta I}{P2}.MU_2 \leqslant 0 \implies \frac{MU_1}{P1} = \frac{MU_2}{P2} \implies \frac{MU_1}{MU_2} = \frac{P1}{P2}$. Note that this is true only when both x_1 and x_2 are greater than 0.
- Above equation implies that the rate at which you are willing to trade goods (MRS) must equal the the rate at which the market allows you to trade them (Price Ratio)

- $\frac{MU_1}{P1}, \frac{MU_2}{P2}\left(\frac{MU_i}{Pi}, \text{ in case of more than 2 goods}\right)$ are referred to as bang for the buck. It's the marginal utilitiy obtained per dollar spent on the respective (indicated by the subscript) good.
- In the cases of multiple goods, it's possible that this ratio is less for some goods than the others. In all cases where it's lesser, we must spent less money on those goods.
- When x_1 has more utility $\frac{MU_1}{P1} \geqslant \frac{MU_2}{P2}$, and when x_2 has more utility $\frac{MU_2}{P2} \geqslant \frac{MU_1}{P1}$.
- V as a function of $P1, P2 \text{ and } I$ is called *indirect utility function*.
- $\frac{\partial V}{\partial I}$ is called *marginal utility of Income.* Often denoted as λ, this represents the *shadow price* of your budget. It tells you exactly how much extra happiness you would get if someone gave you one more rupee of income.
- $\frac{P1}{P2}$ is called the *relative attractiveness* of one good in terms of the other.
- Examples

  Case 1: $U(x,y) = x^{0.5} + y$, such that $P_x x + P_y y \leqslant I$

  $MU_x = \frac{\partial U}{\partial x} = 0.5x^{-0.5}. \quad MU_y = \frac{\partial U}{\partial y} = 1$

  $$\frac{MU_x}{MU_y} = \frac{P_x}{P_y} \implies \frac{1}{2\sqrt{x}} = \frac{P_x}{P_y} \implies \sqrt{x} = \frac{P_y}{2P_x} \implies x^* = \frac{P_y^2}{4P_x^2}$$

  Substituting x^* back into the budget line, we get $P_x\left(\frac{P_y^2}{4P_x^2}\right) + P_y y = I.$ This implies $\frac{P_y^2}{4P_x} + P_y y = I \implies P_y y = I - \frac{P_y^2}{4P_x} \implies y^* = \frac{I}{P_y} - \frac{P_y}{4P_x}$

  Case 2: $U(x,y) = min(2x, 5y)$, such that $P_x x + P_y y \leqslant I$

  In this case, we can't use calculus, since it's not convex and hence write $2x = 5y$

  $$P_x x + \frac{P_y.2x}{5} = I \implies x\left(\frac{5P_x + 2P_y}{5} = I\right) \implies x^* = \frac{5I}{5P_x + 2P_y}$$

  Since we know $2x = 5y$, we can say $y = \frac{2x}{5} = \frac{2I}{5P_x + 2P_y}$
- Above calculated values of x^* and y^* are called *Marshallian demand functions*.
- When income increases, budget line shifts to the right. Consumption increases for normal good, and decreases for inferior good. Note that both goods can't be inferior.

- If the cost of inputs decreases, the supply function shifts outward, implying that firm can produce more and supply more quantity at the same price. However, the price reduces, firm will supply lesser quantity.
- When the price of a substitute good increases, the supply function shifts left, implying that the firm is now willing to supply lesser. WHen the price of a complement good increases, the supply funciton shifts right, implying that the firm will supply more of it.

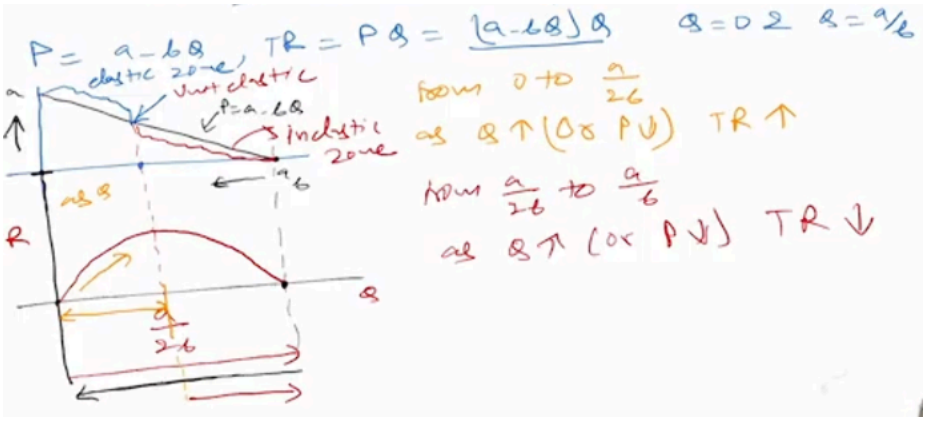
Week6

- Price elasticity of demand is defined as the proportional change in demand ($\triangle Q / Q$) with respect to the proportional change in price($\triangle P / P$). Thus, $\epsilon = \frac{\triangle Q}{\triangle P} * \frac{P}{Q}$. This can be written as $-\frac{dQ}{dP} * \frac{P}{Q} = -\frac{dQ}{Q} \div \frac{dP}{P} = -\frac{d(lnQ)}{d(lnP)}$. Note that elasticity has typically a negative sign.
- Price, Income and Advertisement affects the demand/supply, which means each of the above said attributes have corresponding elasticities.
- If $P = \frac{10}{Q}, Q = \frac{10}{P} \implies logQ = log10 - logP \implies \frac{d(logQ)}{d(logP)} = 0 - 1 = -1$. Thus, elasticity is constant throughout the demand curve. This means that if P changes (increases) by 1%, Q also changes (decreases) by 1%.
- If $P = a - bQ, \frac{dP}{dQ} = -b$. This implies, $\epsilon = -\frac{dQ}{dP} * \frac{P}{Q} = -\frac{1}{b} * \frac{a - bQ}{Q}$. In the demand curve, $\left(\frac{a}{b}, 0\right)$ and $(0, a)$ are the x-intercept and y-intercept respectively. At the x-intercept, $\epsilon = -\frac{1}{b} * \frac{a-a}{a} = 0$. At the y-intercept, $\epsilon = -\frac{1}{b} * \frac{a-0}{0} = -\infty$. Note that the elasticity decreases along the demand curve.
- By convention, we take the mid-point while calculating *arc elasticity*.
- If $\epsilon > 1$, it's considered elastic, else inelastic.
- Total Revenue $TR = PQ \implies \frac{dTR}{dQ} = P + Q.\frac{dP}{dQ} = P\left[1 + \frac{Q}{P}.\frac{dP}{dQ}\right] = P\left[1 - \frac{1}{\epsilon}\right]$.
  - $\epsilon = 1 \implies \frac{dTR}{dQ} = 0$. Unit elastic zone. Here, Total Revenue doesn't change by quantity produced.
  - $\epsilon > 1 \implies \frac{dTR}{dQ}$ is a positive quantity. Elastic zone. Here, if P is decreased by 1%, Q increases by more than 1%. IN this case, total revenue increases. If P is increased by 1%, Q decreases by more than 1%. In this case, total revenue

decreases. Thus, in order to increase total revenue, do not increase price when operating in elastic zone.

- $\epsilon < 1 \Longrightarrow \frac{dTR}{dQ}$ is a negative quantity. Inelastic zone. Here, if P is increased by 1%, Q increases by less than 1%. In this case, total revenue increases. If P is decreased by 1%, Q decreases by less than 1%. In this case, total revenue decreases. Thus, in order to increase total revenue, increase price when operating in inelastic zone.

- Following diagram shows the elastic/inelastic zones of price elasticity clearly.



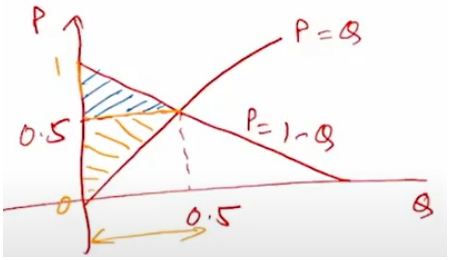
- Cross-elasticity of Demand is defined as the proportional change in demand ($\Delta Q_1 / Q_1$) of one good with respect to the proportional change in price($\Delta P_2 / P_2$) of another good. If the goods are substitutes, this number is positive. However, in the case of complementary goods, this number is negative.
- In the case of normal goods, income elasticity of demand is positive, implying that as independent variable increases, demand increases. In the case of inferior goods, income elasticity of demand is negative, implying that as independent variable increases, demand decreases.
- Price elasticity of supply is defined as the proportional change in supply ($\Delta Q^s / Q^s$) with respect to the proportional change in price($\Delta P / P$). Thus, $\eta = \frac{\Delta Q^s}{\Delta P} * \frac{P}{Q^s}$. This can be written as $-\frac{dQ^s}{dP} * \frac{P}{Q^s} = -\frac{dQ^s}{Q^s} \div \frac{dP}{P} = -\frac{d(lnQ^s)}{d(lnP)}$. Note that elasticity has typically a positive sign.
- Ranges of Elasticity

Elasticity Value	Meaning
$E < 0$	Negative elasticity → demand falls when the variable rises (e.g., own-price elasticity, complements, inferior goods).
$E = 0$	Perfectly inelastic → demand does not respond at all to changes in the variable.
$0 < E < 1$	Inelastic → demand responds less than proportionally.
$E = 1$	Unit elastic → demand changes proportionally to the variable.
$E > 1$	Elastic → demand responds more than proportionally.
$E \to +\infty$	Perfectly elastic → even a tiny change in the variable causes demand to collapse or explode.
$E \to -\infty$	Perfectly elastic but in the opposite direction (theoretically possible if the relationship is strongly negative).

Some more definitions:

  - Own-price elasticity: usually negative, often reported in absolute value.
  - Cross-price elasticity: positive for substitutes, negative for complements.
  - Income elasticity: positive for normal goods, negative for inferior goods.
  - Advertising elasticity: usually positive, but can be negative if advertising backfires.

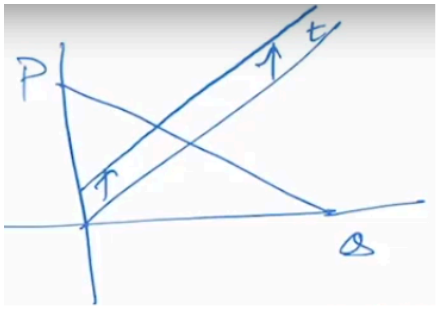
- The point where supply curve meets with the demand curve is called the market equilibrium. The price and quantity at this point decides the consumer/producer suplus. If $P_b = 1 - Q_b$ represents the demand curve and $P_s = Q_s$ represents the supply curve, the market equilibrium point occurs at (0.5, 0.5).



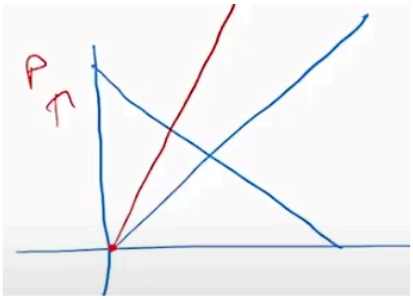
Consumer Surplus (CS) is calculated as the area under the demand curve, bounded by $P = 0.5$. Similarly, Producer Surplus (PS) is calculated as the area above the supply curve, bounded by $P = 0.5$. Thus, $CS = \frac{1}{2} * 0.5 * 0.5 = 0.125$ and $PS = \frac{1}{2} * 0.5 * 0.5 = 0.125$. Finally, $Total\ surplus = CS + PS = 0.5$

- Taxes can be specific (fixed), ad valorem (on-value tax), and lump-sum.

- If the govt. imposes a tax on buyers, the demand curve shifts downward, parallel to the original demand curve. If it imposes on sellers, the supply curve shifts upward, parallel to the original supply curve. The wedge between the two curves (only the shifted part) is the tax.
- $P_b - P_s = tax\ component$
- Typically, buyer and seller shares the govt.taxes equally. However, the division of tax burden depends on the relative elasticities. If demand is inelastic (buyers less sensitive to price), buyers bear more of the tax. If supply is inelastic (sellers less flexible), sellers bear more of the tax. If both are equally elastic, the burden is split evenly.
- 4 cases of incidence of specific taxes
  - Perfectly elastic demand: Seller pays the full share of taxes.
  - Perfectly inelastic demand: Buyer pays the full share of taxes.
  - Perfectly elastic supply: Buyer pays the full share of taxes.
  - Perfectly inelastic supply: Seller pays the full share of taxes.
- Buyer's share of taxes is given by the formula $\frac{\eta_s . t}{\epsilon_d + \eta_s}$. Seller's share of taxes is given by the formula $\frac{\epsilon_d . t}{\epsilon_d + \eta_s}$
- As mentioned before, In the case of specific tax of seller, the supply/demand curve shifts upwards/downwards respectively, parallel to the original curve. In the case of ad valorem tax, the shift is a rotation. This is because, instead of increasing by a constant amount, ad-valorem tax is a function of price - $P(1+t)\ instead\ of\ P+t$



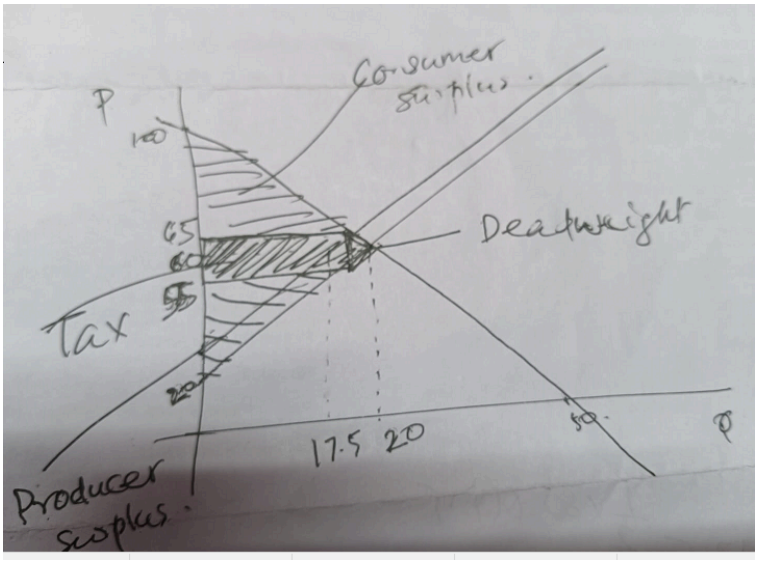
Specific tax



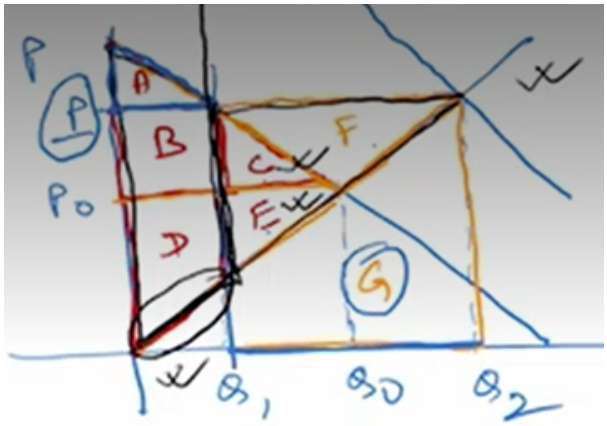
Ad-valorem tax

- As an example,consider the following:
  - Demand: $P = 100 - 2Q$
  - Supply: $P = 20 + 2Q$
  - With no tax, $100 - 2Q = 20 + 2Q \implies Q = 20,\ P = 60.$
  - With 10% tax, new supply curve, $P = 30 + 2Q$. New equilibrium occurs at $Q = 17.5$.
  - Substituting this value of Q into demand curve, $P_b = 65$

– Substituting this value of Q into supply curve, $P_s = 55$
– Thus, buyer pays 65, seller receives 55, govt receives 10 as tax, quantity drops $20 \rightarrow 17.5$, creating a deadweight triangle $\frac{1}{2}.(20 - 17.5) * 10 = 12.5$



- Scenarios of market intervention to raise price.

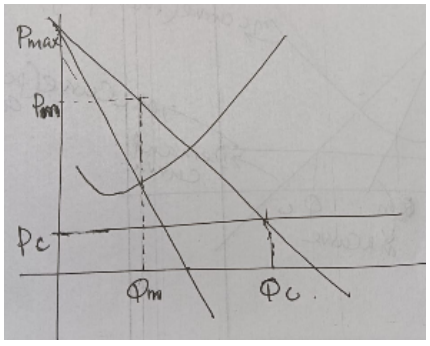


Scenario	CS (Consumer Surplus)	PS (Producer Surplus)	G (Government Expenditure / Revenue)	DWL (Deadweight Loss)	Total Surplus
No Intervention	A + B + C	D + E	0	0	A + B + C + D + E
Price Floor	A	B + D	0	C + E	A + B + D
Price Support	A	B + D + C + E + F	−(C + E + F + G)	C + E + G	A + B + D − G
Production Quota	A	B + D	0	C + E	A + B + D
VPR (Variable Price	A	B + D + C + E + F	−(C + E + F)	C + E	A + B + D

- $Consumer\ surplus = \frac{1}{2} * Q_m * (P_{max} - P_m)$.

  $Deadweight\ loss = \frac{1}{2} * (Q_c - Q_m) * (P_m - P_c)$

  where P_{max} is the choke price (when quantity is set to 0), Q_m is the quantity at $MR = MC$, P_m is the market price at that quantity (projected onto the market demand curve), $P_c = MC$ when MC is the price in a perfectly competitive market and Q_c is the market demand.



Week7

- Techniques used drawing up a demand schedule
  - Managerial Judgment is a way to use accumulated knowledge of sales teams & product teams. It involves coming up with a *Source of Volume* model

(composition of buyers) with justification.
  - Analogous Product. Works well for the *next* generation product, and not for an entirely new product.
  - Benchmarking. Comparision with competitive products. Doesn't work for an entirely new product.
  - Focus groups. Qualitative market research, based on questions posed among one or more discussion groups.
  - Surveys. Survey typical buyers, and uses a 5-point scale.
  - Field experimentation or simulation can be a useful technique.
  - Regression Analysis. Create a model (equation) from past historical data.
  - Conjoint Analysis.
  - Economic Value to Customers. In order to decide Price of product ***a***, choose a reference product ***b***.

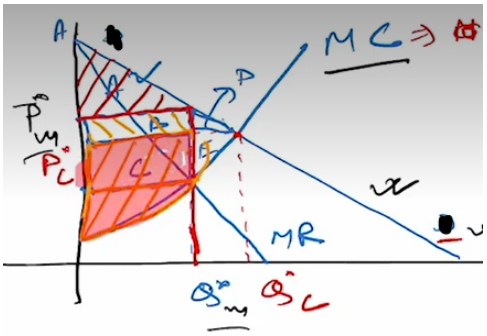
$$Utility_a \geqslant Utility_b \implies Gain_a - Price_a \geqslant Gain_b - Price_b \implies$$
$$Price_a \leqslant (Gain_a - Gain_b) + Price_b \implies Differential\ Gain + Price_b$$

- In perfect competition, the demand curve is downward sloping for the whole market, but for a specific firm, it is a *horizontal line* at the equilibrium price. On the other hand, in monopolistic environment, the demand curve is always downward sloping.
- Though, theoretically, the seller can set a price as he wants to (increasing the margin $P - AC$), the buyer will purchase only less quantity (reducing the volume he buys). This is called *Volume Vs Margin tradeoff*. Thus,
  $Revenue = Margin * Volume = (P - AC) * Q$
- If $P = 10 - Q,\ TR = (10 - Q)Q = 10Q - Q^2$. Since $MR = \frac{\partial TR}{\partial Q},\ MR = 10 - 2Q$.
  $AR = \frac{TR}{Q} = (10 - Q)$.
- Continuing on the same problem, if $TC = 2Q,\ MC = 2$. Now, if we need to maximize profit, $MR = MC \implies 10 - 2Q = 2 \implies Q^* = 4 \implies P^* = 10 - 4 = 6$
- Monopolist will produce less amount and sell at a higher price compared to a perfectly competitive market.
- IN a general setting of a monopoly environment, if we need to maximize the profit, differentiate the expression $P.\ Q - C(Q)$ and equate to zero. This results in the equation
  $\frac{\partial P}{\partial Q}.\ Q + P.1 - MC(Q) = 0.$ Alternatively,
  $P\left[1 + \frac{Q}{P}\frac{\partial P}{\partial Q}\right] - MC(Q) = 0 \implies P\left[1 - \frac{1}{\epsilon}\right] = MC \implies 1 - \frac{1}{\epsilon} = \frac{MC}{P} \implies \mathbf{\frac{1}{\epsilon} = \frac{P - MC}{P}}.$
  This quantity is called markup index.

NOTE: In a perfect competitive environment, P is constant and hence differentiating

the above leads to $P = MC$. In this case, elasticity doesn't come into picture. In other words, it's perfectly elastic($\epsilon = \infty$)

- Note also that in a non-elastic zone ($0 \leqslant \epsilon < 1$), $1 - \frac{1}{\epsilon}$ is negative and hence leads to reducing profit. So, monopolist wouldn't work in this zone.
- Monopoly would result in welfare loss indicated by D+E, as shown in the following figure. Note that the demand curve (top line) and *MC* intersects to give the equilibrium quantity (Q_c^*) in a competitive market. $MR = MC$ results in the quantity (Q_m^*) sold in monopolist market. P_c^* and P_m^* are the corresponding prices.



- Monopolist shouldn't produce anything when Average Cost > Price, else the profit can become negative. In this case, it's better to shutdown production. So, $MR = MC$ is necessary, but not sufficient condition for a monopolist to maximize the profit.
- A monopolist will never produce in the inelastic portion of the demand curve. In the inelastic region, raising price increases total revenue but also reduces quantity demanded only slightly. However, because marginal revenue becomes negative in this region, producing here would reduce profit.

Week8

- Price discrimination
  - 1st degree: Price per unit is different for each buyer and price per unit also may vary with units. This doesn't work practically, because the supplier will not possibly know the demand function for each buyer.
  - 2nd degree: Price per unit is different for each unit, but remains same across buyers (purchasing the same unit)
  - 3rd degree: Price per unit is different for each buyer, but remains same for the same buyer irrespective of how many units he/she buys.
- Following outlines a price discrimination scenario for a monopolist selling to two different markets. Here are the specific examples and calculations used:

Market Data:

Market 1 Demand: $Q1 = 40 - 0.2P1$
Market 2 Demand: $Q2 = 25 - 0.3P2$
Total Cost (TC): $30 * Q$

$$Marginal\ Cost\ (MC) = \frac{d(TC)}{dQ} = 30$$

Case A: Without Price Discrimination (Single Price)
In this scenario, the monopolist charges the same price in both markets.

Total Demand: $Q = (40 - 0.2P1) + (25 - 0.3P2) = 65 - 0.5P$ (ignored subscripts)
Inverse Demand: $P = 130 - 2Q$
Marginal Revenue (MR): $\frac{d(PQ)}{Q} = \frac{d\left(130Q - 2Q^2\right)}{dQ} = 130 - 4Q$
For profit maximization, set MR = MC
$130 - 4Q = 30 \Longrightarrow 4Q = 100 \Longrightarrow Q = 25$
$P = 130 - 2 * 25 = \$80$
$Profit = TR - TC = (\$80 * 25) - (\$30 * 25) = \$2000 - \$750 = \$1250$

Case B: With Price Discrimination (Segmenting)
In this scenario, the monopolist charges different prices based on the market.

Market 1 Calculations:

Demand 1: $Q1 = (40 - 0.2P1)$
Inverse Demand 1: $P1 = 200 - 5Q1$
Marginal Revenue 1: $MR1 = 200 - 10Q1$
For profit maximization, set MR1 = MC
$200 - 10Q1 = 30 \Longrightarrow 10Q1 = 170 \Longrightarrow Q1 = 17$
$P1 = 200 - 5 * 17 = \$115$

Market 2 Calculations:

Demand 2: $Q2 = (25 - 0.3P2)$
Inverse Demand 2: $P2 \approx 83.33 - 3.33Q2$
Marginal Revenue 2: $MR2 \approx 83.33 - 6.66Q2$
For profit maximization, set MR2 = MC
$83.33 - 6.66Q2 = 30 \Longrightarrow Q2 \approx 8$
$P2 = 83.33 - 3.33 * 8 \Longrightarrow P2 \approx \$56.67$

Total Profit Calculation (Case B):

Total Profit: $TP = P1 * Q1 + P2 * Q2 - 30(Q1 + Q2) =$
$\$115 * 17 + \$56.67 * 8 - \$30 * (17 + 8) = \$1955 + \$453.36 - \$750 = \$1658.36$

Note that the total profit in a segmented market is $\approx \$400$ more than single-price market.

- Under first-degree discrimination, you should use $P = MC$, instead of $MR = MC$ (single-price strategy).
- Consider the following setup.
  - $MR = 100 - 2Q. MC = 20.$

    Case 1: Single-Price Monopoly
    For profit maximization MR = MC. $100 - 2Q = 20 \Longrightarrow Q = 40.$
    Since MR is obtained by differentiating Total Revenue (PQ), we could write,

    $$PQ = \int (100 - 2Q)dQ = 100Q - Q^2 \Longrightarrow P = 100 - Q.$$

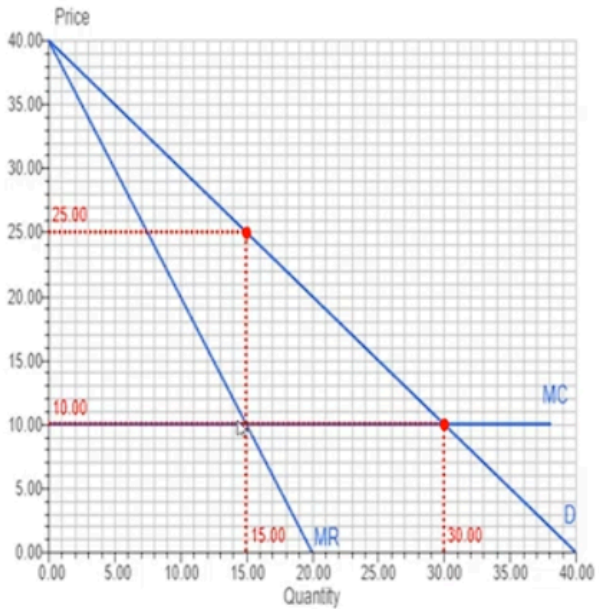
    Thus, in this case,
    $P = 100 - 40 = 60.$ Hence, $Profit\ \pi = (P - MC).\ Q = (60 - 20).40 = 1600.$

    Case 2: First-degree price discrimination.
    For profit maximation P = MC. $100 - Q = 20 \Longrightarrow Q = 80.$
    Since P is obtained by differentiating Total Revenue, we could write,

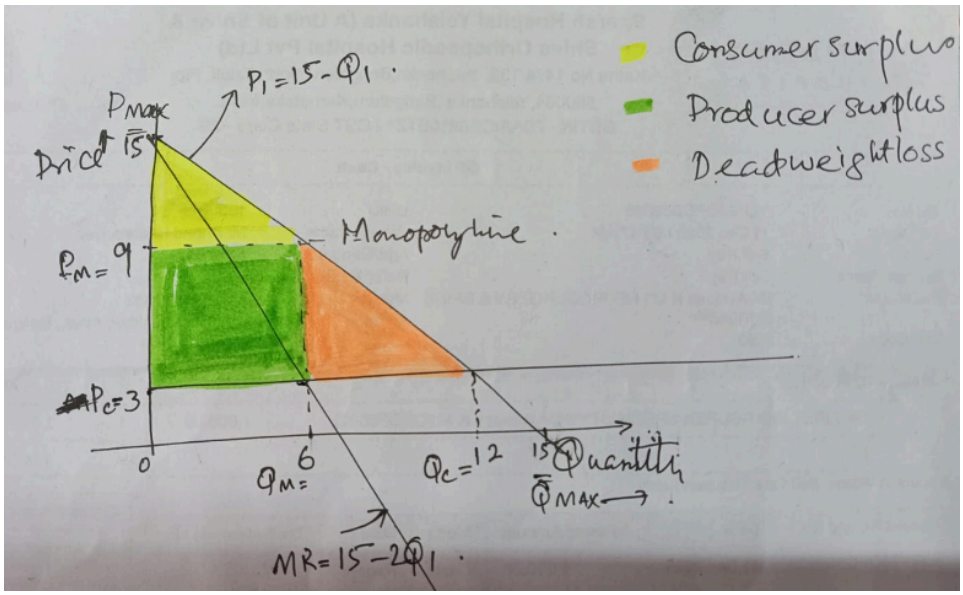
    $$TR = \int_0^{80} (100 - Q)dQ = \left[100Q - \frac{Q^2}{2}\right]_0^{80} \Longrightarrow TR = 8000 - 3200 = 4800.$$

    $TC = MC * Q = 20 * 80 = 1600$
    $Profit\ \pi = TR - TC = 4800 - 1600 = 3200$
- For the above example, calculation of surplus under single-price and 1st degree discrimination are given below.

	Single Price Outcome, P = $25	1st Degree Price Disc.
Cons. Surplus	0.5(40 – 25)(15) = $112.50	$0
Prod. Surplus	(25 – 10)(15) = $225	0.5(40 - 10)(30) = $450
Total Surplus	$112.50 + $225 = $337.50	$450

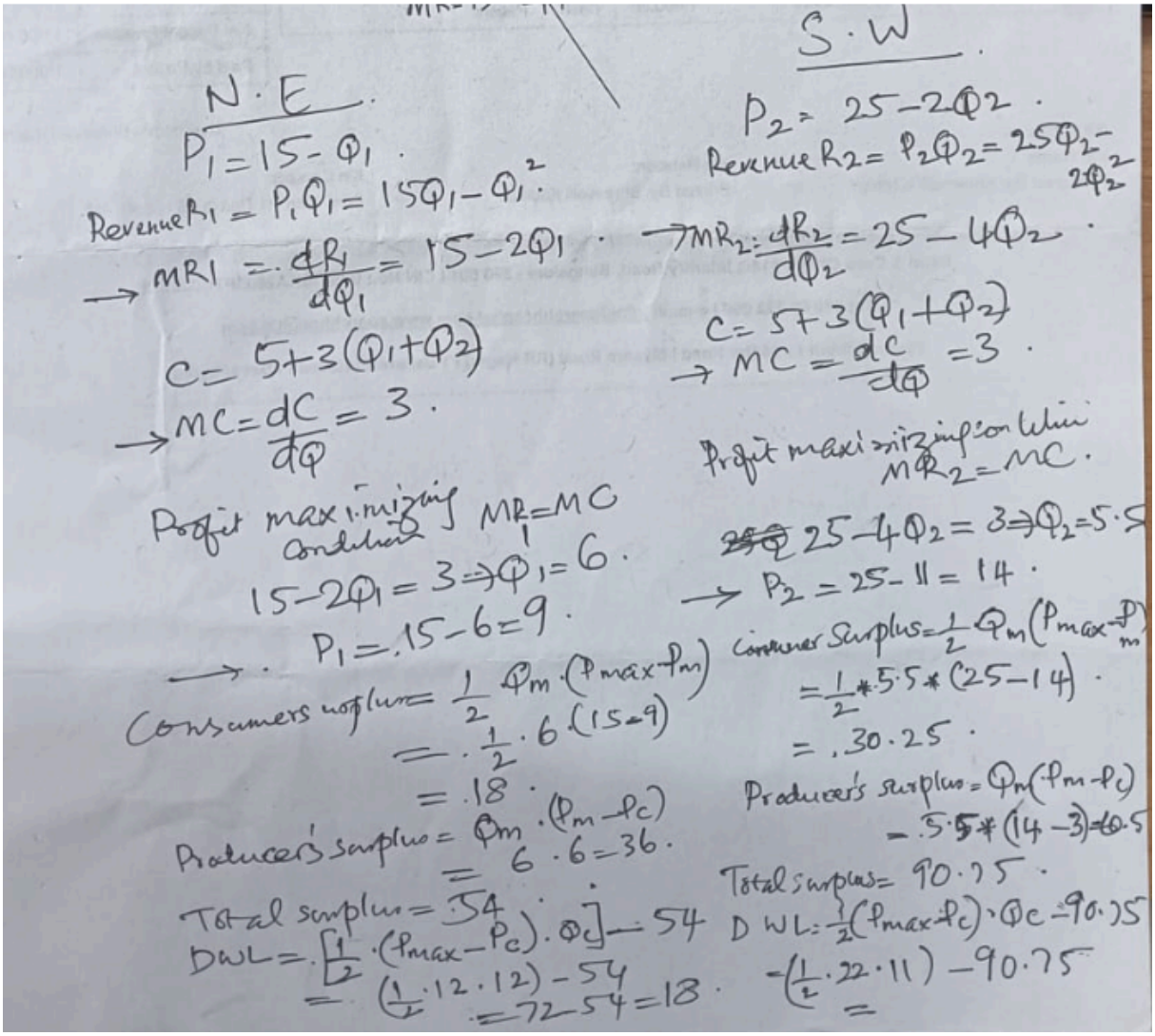


- Problem from GA-8
  A monopolist is deciding how to allocate output between two geographically separated markets (North-East and South-west). Demand curves for the two markets are $P_1 = 15 - Q_1$ and $P_2 = 25 - 2Q_2$ respectively. The monopolist's total cost is $C = 5 + 3(Q_1 + Q_2)$. Suppose that the monopolist can price discriminate and answer the following questions:



The above diagram shows the demand curve in North East market.

- Given below are the calculations (including the surplus), for both markets.



- While a monopolist is the only seller of a product, a monopsonist is the single buyer of an input, such as labor in a company town.
- A monopsonist acts as a price maker in the input market and a price taker in the output market
- The profit-maximizing quantity of labor (L*) is determined by equating the marginal revenue of the input with the marginal cost of hiring it.
- The monopsony model is mathematically similar to a monopoly, with the key difference being that the monopsonist operates on the supply side of the input market rather than the demand side of the output market.
- Profit in the input market can be expressed as $P * f(L) - w(L) * L$, where P is the

price, f(L) is the production function, w(L) is the input cost and L is the number of labor units.

- For the profit maximization, we use the first-order condition

$$P * f'(L) = w(L) + w'(L) * L = w(L)\,[1 + w'(L) * L / w(L)] = w[1 + w' * L / w] = w[1 + 1/\eta]$$

NOTE: η represents the price elasticity of supply. Remember, price in an input market is wages.

- If $w = a + bL$ represents the supply function, we can find the marginal cost by multiplying this by number of labor units ($wL = aL + bL^2$) and differentiating with respect to L ($a + 2bL$)
- 4CR identifies the top 4 firms that has highest sales revenue, production output, employees and capacity. $4CR = \frac{Sales\ of\ 4\ largest\ firms}{Total\ industry\ sales}$. Helps identify market structure.

Market Structure	Typical 4CR Range	Characteristics
Perfect Competition	0% (or very close to 0)	Thousands of tiny firms, no single firm has measurable share, homogeneous products, free entry/exit
Monopolistic Competition	Low (0–40%)	Many firms, each with small share, differentiated products, easy entry/exit
Oligopoly	Moderate to High (40–100%)	Few large firms dominate, significant barriers to entry, strategic interdependence, possible collusion
Monopoly	100%	One firm controls the entire market, no close substitutes, very high barriers to entry

NOTE: 4CR ranges from 0 to 100.

- Herfindahl Hirschman Index (HHI) measures sum of the square of market sales percentage across all firms. It can be expressed as $\sum_{i=1}^{n} s_i^2$, where n represents the number of firms and s_i represents the market sales percentage of the ith firm.

Quiz-2

- In quasi-linear utility functions, the marginal utility of income remains constant.
- Quasi-linear preferences are non-linear in one good, and linear in the other.
- Marginal utility changes for a monotonic transformation of the utility function
- MRS doesn't change for a monotonic transformation of utility function
- The fraction of quantity tax paid by buyers rises as supply becomes more own-price elastic, and as demand becomes less own-price elastic

- The fraction of quantity tax paid by buyers reduces as supply becomes less own-price elastic, and as demand becomes more own-price elastic.
- A profit-maximizing monopoly will produce that output for which marginal cost equals marginal revenue
- A natural monopoly occurs when one firm can supply the entire market more cheaply than can a number of firms
- Monopolist never produces in the inelastic region of the demand curve

- Formulae
  - $\frac{P - MC}{P} = \frac{1}{\epsilon_d}$
  - For a monopsonist, use $ME = Demand$ to find the optimal quantity q. $ME = \frac{dTE}{dQ}$. Substitute q in supply curve, to find the price. Note: ME is marginal expenditure.
  - Producer surplus = $\int_0^q (P - MC).\, dq$
  - For a monopolist, use $MR = MC$ to find the optimal quantity. Substitute this quantity in demand curve, to find the price.
  - $Consumer\ surplus = \frac{1}{2} * Q_m * (P_{max} - P_m).$
  - $Deadweight\ loss = \frac{1}{2} * (Q_c - Q_m) * (P_m - P_c)$
  - $Optimality\ condition: MRS = price\ ratio \implies \frac{MU_x}{MU_y} = \frac{P_x}{P_y}.$

NOTE: When solved, if the above equation yields x or y to be negative, it's not possible solution. In that case, take x (or y) = 0, and calculate the y (or x)

Week9

- In cournot duopoly, optimal quantity is in between the optimal quantity for monopoly and perfect competition. $Q_{PCM} > Q_{CD} > Q_M$. Similarly, optimal price is in between the optimal quantity for monopoly and perfect competition. $P_M > P_{CD} > P_{PCM}$.
- Problem in Cournot duopoly
-